Dataset Suggestions for Final Project

This document provides curated, accessible datasets for your final project. Choose data that matches your statistical method and research question.

General Dataset Repositories

1. Kaggle Datasets

- **URL:** https://www.kaggle.com/datasets
- **Access:** Free account required
- **Best for:** Beginner-friendly, clean datasets with documentation
- **File formats:** CSV, JSON
- **Suggested search terms:** “health”, “education”, “survey”, “correlation”, “categorical”

2. UCI Machine Learning Repository

- **URL:** https://archive.ics.uci.edu/ml/datasets.php
- **Access:** Free, no account needed
- **Best for:** Classic datasets, well-documented
- **File formats:** CSV, TXT, ARFF
- **Note:** Great for regression and classification projects

3. Data.gov

- **URL:** https://data.gov
- **Access:** Free, no account needed
- **Best for:** Government data, public health, demographics
- **File formats:** CSV, JSON, XML

- **Note:** Very large datasets available - you may want to sample

4. Google Dataset Search

- **URL:** https://datasetsearch.research.google.com
- **Access:** Free search engine for datasets
- **Best for:** Finding datasets across multiple sources
- **Note:** Check license and accessibility before committing

5. FiveThirtyEight Data

- **URL:** https://data.fivethirtyeight.com
- **Access:** Free, hosted on GitHub
- **Best for:** Well-documented, interesting datasets with context
- **File formats:** CSV
- **Topics:** Politics, sports, economics, culture

Datasets by Statistical Method

Chi-Square Test (Categorical Data)

Titanic Survival Dataset

- **Source:** Kaggle
- **URL:** https://www.kaggle.com/competitions/titanic/data
- **Variables:** Survival, passenger class, sex, age, fare
- **Sample size:** ~900 passengers
- **Research questions:** Is survival independent of passenger class? Gender?
- **Notes:** Classic dataset, very well-documented

Movie Genre and Rating

- **Source:** Kaggle (search “IMDB movies”)
- **Variables:** Genre, rating, year, duration
- **Sample size:** Varies (1,000-50,000+)
- **Research questions:** Is genre independent of rating category? Year?

Student Survey Data

- **Source:** UCI Repository
- **URL:** https://archive.ics.uci.edu/ml/datasets/Student+Performance
- **Variables:** School, sex, study time, grades, alcohol consumption
- **Sample size:** 649 students
- **Research questions:** Is alcohol consumption independent of grades?

Correlation Analysis (Two Continuous Variables)

Life Expectancy and GDP

- **Source:** Gapminder / Kaggle
- **URL:** Search “gapminder” on Kaggle
- **Variables:** Life expectancy, GDP per capita, population, year
- **Sample size:** 200+ countries × multiple years
- **Research questions:** Correlation between GDP and life expectancy?

Air Quality Data

- **Source:** UCI Repository
- **URL:** https://archive.ics.uci.edu/ml/datasets/Air+Quality
- **Variables:** Temperature, humidity, pollutant levels
- **Sample size:** 9,000+ hourly measurements
- **Research questions:** Correlation between temperature and ozone?

Housing Prices

- **Source:** Kaggle (search “housing prices”)
- **Variables:** Price, square footage, bedrooms, location
- **Sample size:** Varies (500-20,000+)
- **Research questions:** Correlation between square footage and price?

Exam Scores Dataset

- **Source:** Kaggle (search “students performance”)
- **Variables:** Math score, reading score, writing score, study time
- **Sample size:** 1,000 students
- **Research questions:** Correlation between study time and scores?

Linear Regression (One Outcome, One or More Predictors)

Boston Housing

- **Source:** Kaggle
- **URL:** Search “Boston housing dataset”
- **Variables:** Median home value, rooms, crime rate, distance to employment
- **Sample size:** 506 neighborhoods
- **Research questions:** Predict home value from number of rooms
- **Notes:** Classic dataset for regression

Medical Expenses

- **Source:** Kaggle
- **URL:** Search “insurance medical costs”
- **Variables:** Charges, age, BMI, children, smoker status, region
- **Sample size:** 1,300+ individuals
- **Research questions:** Predict medical costs from age and BMI

Salary Prediction

- **Source:** Kaggle (search “salary data”)
- **Variables:** Salary, years of experience, education, job title
- **Sample size:** Varies
- **Research questions:** Predict salary from experience and education

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t-Test (Compare Two Groups)

Sleep Efficiency

- **Source:** Kaggle (search “sleep health”)
- **Variables:** Sleep duration, age, gender, occupation, sleep quality
- **Sample size:** 400+ individuals
- **Research questions:** Do males and females differ in sleep duration?

Plant Growth Experiments

- **Source:** Built into R (datasets package)
- **R command:** `data(PlantGrowth)`
- **Variables:** Plant weight, treatment group
- **Sample size:** 30 plants
- **Research questions:** Do treatment groups differ from control?

Coffee and Alertness

- **Source:** Kaggle (search “coffee” or create simulated data)
- **Variables:** Alertness score, coffee consumption (yes/no), age
- **Sample size:** Varies or simulate ~100
- **Research questions:** Does coffee consumption affect alertness?

Test Scores: Method Comparison

- **Source:** Create or simulate
- **Variables:** Test score, teaching method (traditional vs. flipped)

- **Sample size:** 50-100 students
- **Research questions:** Do teaching methods produce different scores?

ANOVA (Compare Three or More Groups)

Iris Dataset

- **Source:** Built into R (datasets package)
- **R command:** `data(iris)`
- **Variables:** Sepal length/width, petal length/width, species (3 types)
- **Sample size:** 150 flowers
- **Research questions:** Do petal lengths differ across species?
- **Notes:** Perfect for learning ANOVA

Diet and Weight Loss

- **Source:** Kaggle (search “diet weight loss”)
- **Variables:** Weight loss, diet type (3-4 types), age, gender
- **Sample size:** 100-300 participants
- **Research questions:** Which diet produces most weight loss?

Anxiety by Study Method

- **Source:** Create or Kaggle (search “student anxiety”)
- **Variables:** Anxiety score, study method (individual, group, online)
- **Sample size:** 60-150 students
- **Research questions:** Does anxiety differ by study method?

Bootstrap Confidence Intervals

Bootstrap can be applied to almost any dataset where you want to estimate uncertainty. Good candidates:

Small Sample Datasets

- **Source:** Any dataset above with n < 50
- **Why:** Bootstrap is especially useful for small samples
- **Example:** Estimate mean with 95% CI using bootstrap

Skewed Data

- **Source:** Income data, housing prices, medical costs
- **Why:** Bootstrap doesn't assume normality
- **Example:** Median income with bootstrap CI

Correlation CI

- **Source:** Any correlation dataset above
- **Why:** Bootstrap provides CI for correlations
- **Example:** "What's the 95% CI for correlation between X and Y?"

Creating Your Own Dataset

If you can't find suitable data, you can:

1. Survey Your Classmates

- Quick Google Form with 5-10 questions
- Aim for 30+ responses
- Topics: sleep, study habits, stress, social media use
- **Approval:** Check with instructor first

2. Collect Observational Data

- Count events (e.g., bike vs. car traffic at different times)
- Measure repeated observations (e.g., coffee prices at different shops)

- Record publicly available information (e.g., restaurant ratings)

3. Use Simulation in R

- Generate data that follows your research question
- Useful for understanding method mechanics
- Example: Simulate two groups with known difference for t-test

```
set.seed(123)
control <- rnorm(30, mean=100, sd=15)
treatment <- rnorm(30, mean=110, sd=15)
data <- data.frame(
  score = c(control, treatment),
  group = rep(c("Control", "Treatment"), each=30)
)
```

Dataset Selection Checklist

Before committing to a dataset, verify:

- [ ] **Size:** At least 30 observations (more is better)
- [ ] **Variables:** Contains the types needed for your method
- [ ] **Access:** You can download it easily
- [ ] **Quality:** No major missing data issues (or plan to handle them)
- [ ] **Documentation:** You understand what variables mean
- [ ] **Ethics:** Publicly available or you have permission to use
- [ ] **Interest:** You find the topic engaging
- [ ] **Scope:** Not too complex for a 10-minute presentation

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Tips for Data Exploration

Once you have your data:

1. **Load and inspect -** Look at first few rows, check variable types
2. **Summary statistics -** Means, SDs, ranges, counts
3. **Check for missing data -** Decide how to handle
4. **Visualize -** Histograms, boxplots, scatterplots
5. **Check assumptions -** Before running your statistical test
6. **Ask AI for help -** Use AI to explore and understand your data

Getting Help

If you're struggling to find appropriate data:

- **Office Hours:** Dr. Molitor can suggest datasets for your question
- **Canvas Discussion:** Post what you're looking for, classmates may help
- **AI Tools:** Ask ChatGPT/Claude "Where can I find data about X for Y analysis?"
- **Librarian:** OSU librarians can help locate datasets

Example Dataset-Method Pairings

Research Question	Dataset	Method	Difficulty
Do survival rates differ by passenger class?	Titanic	Chi-Square	Easy
Is there correlation between GDP and life expectancy?	Gapminder	Correlation	Easy

Research Question	Dataset	Method	Difficulty
Can we predict insurance costs from age and BMI?	Medical Insurance	Regression	Medium
Do males and females differ in sleep duration?	Sleep Health	t-test	Easy
Do iris species differ in petal length?	Iris	ANOVA	Easy
What's the 95% CI for median delivery time?	Any small dataset	Bootstrap	Medium

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Dataset Ethics Reminder

- **Use publicly available data** or data you collected ethically
- **Cite your data source** in your presentation and write-up
- **Don't use proprietary data** without permission
- **Respect privacy** - no personal/identifiable information
- **Be transparent** about data cleaning or filtering choices

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Last Updated: Fall 2025

Questions? Post in Canvas discussion or visit office hours